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Key Points:

- We incorporate X-ray imaged fracture apertures, obtained at high-stress in-situ conditions, into 3D DFN flow and transport simulations
- We observe the effects of incorporating real aperture data for fractures on network-scale flow and transport properties for the first time
- Inclusion of aperture variability leads to scale-bridging, where this small-scale feature reorganizes the flow field at the network scale

Supporting Information:

Supporting Information may be found in the online version of this article.

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Scale-Bridging in Three-Dimensional Fracture Networks: Characterizing the Effects of Variable Fracture Apertures on Network-Scale Flow Channelization

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Abstract We incorporate observations of real fracture aperture variability observed in laboratory experiments into an ensemble of three-dimensional discrete fracture network (DFN) simulations to characterize how variations of this micro-scale feature can influence flow and transport behavior at the network scale. A shear fracture is created within a Marcellus shale sample, and the fracture aperture is measured using a triaxial direct-shear device coupled with real-time X-ray imaging at in-situ stress conditions. We construct an ensemble of fracture networks based on natural fractures in Marcellus shale and project regions of the experimental aperture field onto each fracture in the networks. Our calculations demonstrate that the degree of flow channelization, a network-scale flow field structure, is dramatically increased by local changes in the aperture field that in turn affects flow and transport properties.

Plain Language Summary Fluid flow and solute transport through subsurface fractured media are inherently multi-scale phenomena. Individual fractures connect to form complicated networks where relevant length scales span multiple orders of magnitude. In this letter, we address a long-standing question in this research area regarding the relative impact of micro-scale heterogeneity of fracture roughness, that is, internal aperture variability, on flow and transport at the network macro-scale. To do so, we developed a methodology to incorporate real apertures obtained from in-situ observations of dynamically fractured laboratory specimens into three-dimensional discrete fracture network simulations for the first time. This methodology allows us to characterize how variations in fracture aperture can lead to increased flow channelization at the network scale, a phenomenon that we refer to as scale-bridging. Our results show that this natural aperture variability not only modifies the local flow field within a single fracture but can re-structure the global flow field of the entire network. In turn, the distribution of solute transport behavior is drastically modified, the networks' active surface area is drastically decreased, and the degree of flow channelization is markedly increased compared to reference networks that use equivalent smooth fractures.

1. Introduction

The flow of fluids and the associated transport of dissolved chemicals through low-permeability fractured media in the subsurface is primarily confined to fractures and the interconnected networks that they form (Berkowitz, 2002; National Research Council, 1996; Neuman, 2005). A distinguishing geophysical feature of fractured media is the multi-scale heterogeneity that occurs within the system. Although networks are composed of individual fractures, they exhibit fundamentally different behavior than the individual units of which they are comprised. In a single fracture, flow field organization is predominantly determined by surface roughness, aperture and contact area, which result from a variety of different geophysical processes such as mechanical deformation and/or chemical reactions (Candela et al., 2012; Chaudhuri et al., 2009; Detwiler & Rajaram, 2007; Deng & Spycher, 2019; Elkhoury et al., 2013; Frash et al., 2019; Lang et al., 2015; Pyrak-Nolte et al., 1997; Sagy et al., 2007). In contrast, networks exhibit multi-scale structural heterogeneity where relevant length scales span that of individual fracture surfaces through complex interconnections of heterogeneous arrays of fractures up to lithospheric-scale fluid pathways (Bonnet et al., 2001; Davy et al., 2010; Hyman, Dentz, et al., 2019; Hyman, Rajaram, et al., 2019; Lei & Gao, 2018;

Moon et al., 2020; Neuman, 2008; Sweeney & Hyman, 2020; Zoback, 1992; Zoback et al., 1989). While it is generally accepted network structure is a dominant feature controlling flow and transport properties (Hyman & Jiménez-Martínez, 2018; Maillot et al., 2016), the hierarchy of influence amongst length scales is poorly understood.

One of the outstanding questions in this context is how the micro-scale heterogeneity of fracture aperture variability, that is, internal aperture variability, influences flow and transport at the network macro-scale. While aperture variability is known to lead to local flow dispersion and channelization within single fractures (Boutt et al., 2006; Brown, 1987a; Brown et al., 1998; Cardenas et al., 2007; Detwiler et al., 2000; Kang et al., 2016; Renshaw, 1995; Zimmerman et al., 1991; Zou et al., 2015, 2017a, b), it has yet to be definitely established whether and how these micro-scale flow effects impact macro-scale flow observables. It is difficult to carry out the requisite experiments of flow through real rock fracture networks at in-situ conditions within a laboratory to investigate these connections and previous numerical simulations have been limited to using statistical representations of the aperture field within each fracture (de Dreuzy et al., 2012; Frampton et al., 2019; Huang et al., 2019; Karra et al., 2015; Makedonska et al., 2016; Zou & Cvetkovic, 2020). Although some measurements of surface roughness of exhumed and laboratory-manufactured fracture surfaces possess self-affine statistical properties (Brown, 1987b; Lee et al., 1990; Renard et al., 2006; Stigsson & Ivars, 2019), it is unclear if these stochastic representations adequately capture the key attributes (including aperture and contact area) that influence flow at the network scale. Recent advancements in our computational abilities provide a framework to test hypotheses about the importance of one scale of heterogeneity against another. The ability of discrete fracture network (DFN) models to include multi-scale geophysical attributes into a single simulation allows us to probe their relevant impact on flow and transport observations and discover a hierarchy of effects on the flow field and associated solute transport. In turn, DFN simulations have the potential to inform field site operators and decision makers about the worth of particular data and how it could benefit constraining forward models (Zou & Cvetkovic, 2021). Advancing our understanding of how aperture variability influences the structure of the flow field within networks is critical for analysis and forward modeling in variety of civil and industrial applications including enhanced geothermal energy (Frash et al., 2021; Wu et al., 2021), unconventional hydrocarbon extraction from shale formations (Hyman, Jiménez-Martínez, et al., 2016; Middleton et al., 2017), CO_2 sequestration (Jenkins et al., 2015), the long term storage of spent nuclear fuel (Birkholzer et al., 2012; Follin et al., 2014; Joyce et al., 2014; Selroos et al., 2002), ground water aquifer management (Kueper & McWhorter, 1991), and the environmental restoration of contaminated sites (VanderKwaak & Sudicky, 1996).

In this letter, we incorporate observations of real fracture apertures observed in laboratory experiments into an ensemble of three-dimensional DFN simulations. We determine how variations of this micro-scale feature can influence flow and transport behavior at the network scale. We refer to this influence of a smaller scale on the behavior of a large scale as *scale-bridging*, which is distinct from upscaling sub-grid scale attributes into bulk effective parameters. We develop a methodology that allows us to incorporate real aperture roughness obtained from observations of dynamically fractured laboratory specimens on network-scale flow and transport properties for the first time. Shear fractures were created in laboratory specimens of Marcellus shale using a triaxial direct-shear method. We construct an ensemble of networks whose fracture aperture fields are extracted from three-dimensional x-ray microtomographic images of the fractured shale specimens at subsurface stress conditions. Regions of the aperture field from the experimental data are projected onto the fractures in the network so that every fracture in the network has a unique and variable aperture distribution based on the *in-situ* observations. Our results show that this aperture variability not only modifies the local flow field within a single fracture but can re-structure the network-scale flow field of the entire network. These changes in the local flow field alter the distribution of solute transit times through the network and particle pathline lengths, which thereby alters retention processes of a chemical species passing through the network. Moreover, the active surface area of the networks (regions with a significant amount of flow) is drastically decreased compared to reference networks that use equivalent smooth fractures, and the degree of flow channelization is markedly increased. While these simulations are performed at a small scale (cm), it is expected that the patterns observed here would also occur at the field scale, albeit with proportionally larger variations in aperture.

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2. Methodology

2.1. Laboratory Experiments

Shear fracture aperture data were measured using a triaxial direct-shear system coupled with real-time x-ray imaging at in-situ stress conditions (Frash et al., 2017, 2019). This method was accompanied by conventional triaxial compression tests to measure the mechanical properties of the rock (Table S1). The rock used in this study was Marcellus shale from an 'as-is' outcrop in Bedford, Pennsylvania. XRD analysis shows a composition of 66% carbonate, 19% quartz and feldspar, 13% clay, and 2% other, giving this shale a stiff and brittle behavior compared to more clay and organic-rich alternatives. The aperture data used in this study were obtained from a specimen that was triaxial direct-shear fractured at an effective confining stress of 4.4 MPa. The mean pore pressure applied to the sample was 0.6 ± 0.4 MPa. The specimen was prepared as a right-regular cylinder with a length of 25.4 mm and a diameter of 25.2 mm according to (ASTM, 2008). The specimen was initially intact with no notable flaws or pre-existing fractures.

Direct-shear fracturing was completed in an approximately constant strain-rate procedure followed by controlled displacement steps with 2-hr X-ray computed tomography scans between each step. As such, the 3D segmented apertures used in this study are the observed apertures at in-situ stress conditions. The apertures were segmented from the reconstructed 3D x-ray gray-scale data using direct thresholding after applying ring-removal filters (e.g., low-pass ring fraction, 3-point multiple-pass adaptive FIR, and median filters) and bilinear factor of 2× compression to smooth image noise (Figure 1a). Thresholding values were manually selected to isolate the rock and fracture void space by visual comparison to the initial data set. The voxel size was 46.6 ± 0.4 μm after segmentation. The data are available on the digital rocks portal (Frash & Carey, 2018). We consider one fracture created at a shear displacement of 0.60 mm. A reconstructed three-dimensional representation of the fracture is shown in Figure 1b where gray-scale indicates the height of the fractures perpendicular to the principal formation axis, for example, normal distance from the nominal fracture plane. The integrated two-dimensional aperture field is shown in Figure 1c and the resulting aperture distribution is shown in Figure 1d. All apertures below the voxel size (46.6 ± 0.4 μm) are assigned an aperture equal to the voxel size as they are below our resolution; this portion makes up around 22% of the sample surface area. Above the resolution threshold, the apertures appear to follow a normal distribution. We quantify the variable aperture field using the following statistical attributes: The arithmetic mean of the two-dimensional aperture field is $2.38 \cdot 10^{-4}$ m, the geometric mean is $1.85 \cdot 10^{-4}$ m, the log-variance is $1.21 \cdot 10^{-1}$ m^2, with a coefficient of variation of 0.59, and correlation length of $\approx 4 \cdot 10^{-3}$ m.

2.2. DFN Simulations

We incorporate these variable fracture apertures into a set of 3D DFNs whose parameters are loosely based on observed fractured zones in Marcellus shale (Gale et al., 2014; Kavousi et al., 2017; Pankaj et al., 2018; Wang & Carr, 2013) in alignment with the experiments described above, but the networks do not represent a particular site. Flow through such micro-fracture networks are important in shale formations, which occur at these length scales, and are believed to control hydrocarbon production in unconventional reservoirs and changes of permeability in damage zones around faults (Gomila et al., 2021; Hyman, Jiménez-Martínez, et al., 2016; Middleton et al., 2017; Mitchell & Faulkner, 2009; Romano et al., 2020). DFN models explicitly represent fractures as 2D planar objects in 3D space and typically model apertures as perfectly smooth parallel plates. Recent advances in DFN models, where fractures and fracture networks are explicitly represented as opposed to continuum models where their properties are upscaled into effective properties (Karimi-Fard et al., 2006; Neuman, 2005; Oda, 1985; Sweeney et al., 2020), now allow for fracture aperture variability to be incorporated in network-scale simulations (de Dreuzy et al., 2012; Frampton et al., 2019; Huang et al., 2019; Karra et al., 2015; Makedonska et al., 2016; Zou & Cvetkovic, 2020). Using the dfnWorks modeling suite (Hyman, Karra, et al., 2015), we generate 10 separate micro-scale 3D DFNs, each within a centimeter cube. One sample network is shown in Figure 2 (left). The networks are composed of three fracture families whose mean orientations align with the Cartesian axis based on a Fisher distribution (Wood, 1994) with a parameter of 100; there is little variability from the mean orientation. All micro-fractures are squares with sides of length $2 \cdot 10^{-3}$ m, and their locations are uniformly distributed throughout the domain. We select uniform fracture sizes to isolate the effects of variable aperture from that of fracture length and fracture

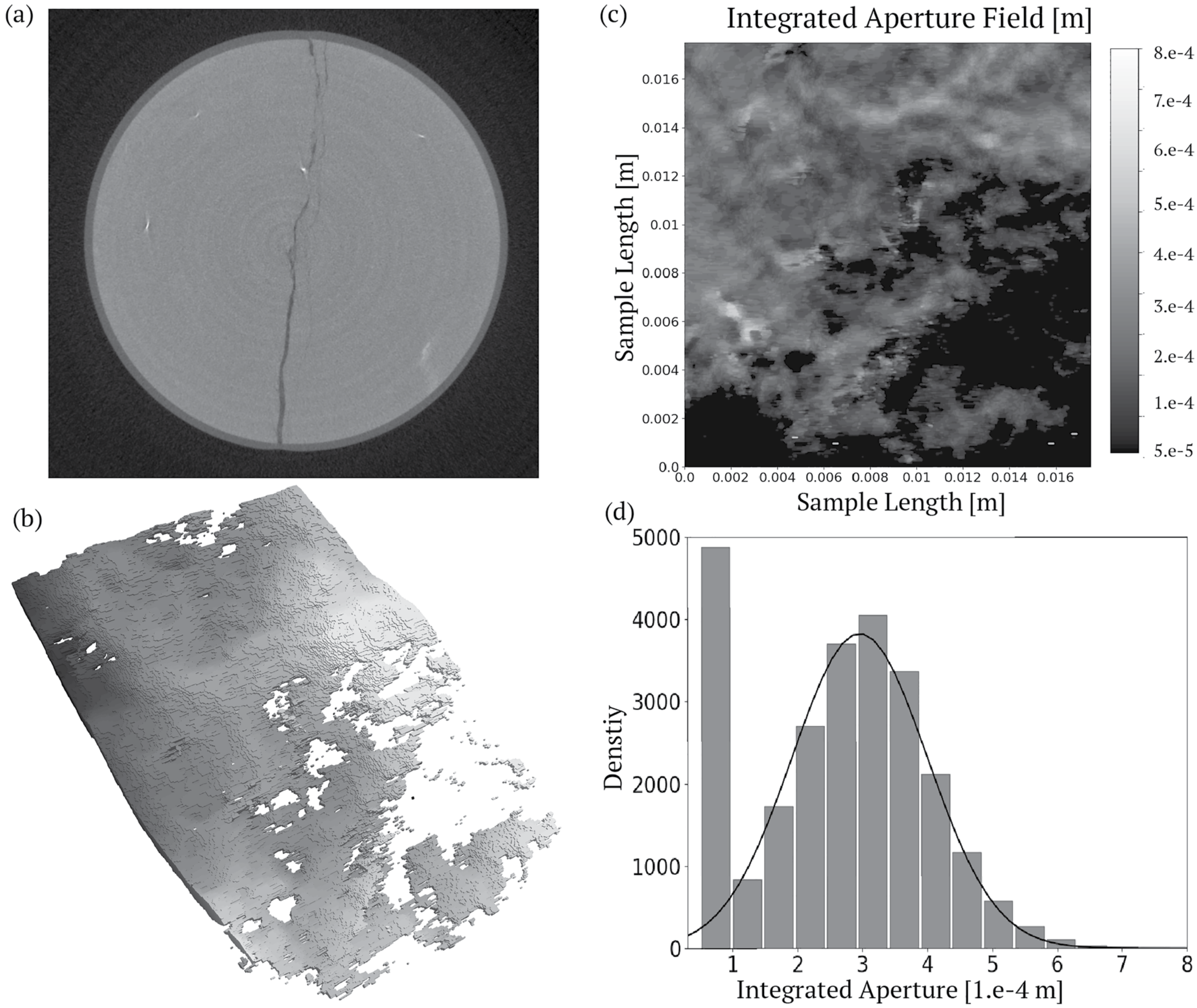

Figure 1. (a) X-ray tomography image of the top of a triaxial direct-sheared sample of Marcellus shale, a right-regular cylinder with a length of 25.4 mm and a diameter of 25.2. (b) Three-dimensional reconstruction of the fracture. Gray-scale represents the height of the fractures perpendicular to the principal formation axis, for example, normal distance from the nominal fracture plane. (c) Integrated two-dimensional fracture aperture. (d) Aperture distribution. All apertures below 50 μm are assigned a value of 50 μm as they are below our resolution.

mean aperture variability, both of which have been extensively studied (Bogdanov et al., 2007; de Dreuzy et al., 2001, 2002; 2004, 2012; Frampton & Cvetkovic, 2010; Huseby et al., 2001; Hyman, Aldrich, et al., 2016; Hyman & Jiménez-Martínez, 2018; Hyman, Dentz, et al., 2019; Joyce et al., 2014; Sherman et al., 2020; Sweeney & Hyman, 2020; Tsang & Neretnieks, 1998; Wellman et al., 2009). We generate 10 independent and identically distributed (i.i.d.) network samples for this study. Four hundred fractures are randomly placed into the domain with equal probability from each family to create a connected network between inflow and outflow boundaries with multiple paths. The number of fractures placed into the domain is a modeling choice so we can test if local variations in aperture impact network-scale flow and transport, and is not based on a particular specimen. However, high density networks are found in damage zones surrounding faults (Gomila et al., 2021; Mitchell & Faulkner, 2009; Romano et al., 2020). Isolated fractures are removed before meshing and the final connected networks contain around 240 fractures and multiple paths connecting the boundaries.

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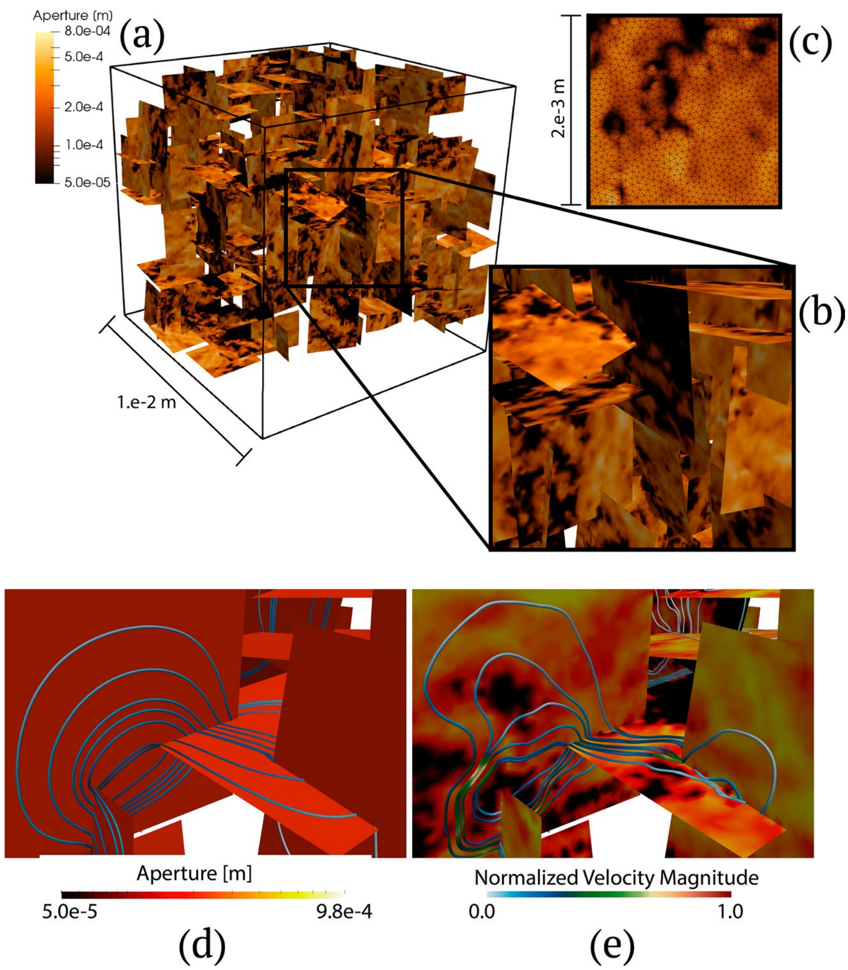

Figure 2. Three-dimensional discrete fracture network model within a centimeter cube (a). Regions of the aperture field from the experimental data from Figure 1c are projected onto the fractures in the network (b). The region is randomly rotated and reflected so that every fracture has a unique variable aperture field. Each fracture is meshed with a conforming Delaunay triangulation at a uniform resolution (c). The effect of the internal variability is shown in (d) and (e) which provides a comparison of pathlines in the same fracture within the same network ((d) Constant aperture (e) Variable aperture). Pathlines are colored by the normalized velocity, values are divided by the maximum. Note that particles have more tortuous pathlines in the variable aperture case as they avoid regions of low-aperture within the fracture planes.

Each fracture is meshed with a conforming Delaunay triangulation at a uniform resolution of edge length 50 μm (Hyman et al., 2014), Figure 2 upper right, which is equivalent to the experimental tomography resolution. Every node in the mesh is assigned a unique aperture value using a randomly selected projection of a different region of the integrated aperture field shown in Figure 1c. This region is randomly rotated and reflected so that every fracture has a unique texture, cf. Figure 2 – lower right. Apertures are independent between fractures and do not correlate across intersections. This methodology ensures that no features are smoothed due to a coarsened mesh nor artificially created below the tomography resolution. We directly use the mechanical aperture as the hydraulic aperture and refer readers to Renshaw (1995) for a discussion about the relationship between the two. Additional details of meshing and the projection method are provided in the Supporting Information S1. We generate 10 variable aperture versions of each of the 10 sample networks, which are also i.i.d. samples. A network with constant aperture, $2.2 \cdot 10^{-4}$ m, which is in between the geometric and arithmetic average of the aperture field, which we refer to as the constant aperture case, is also created for comparison.

We determine the steady-state pressure field and volumetric flow rates of an incompressible isothermal fluid within each network by applying a pressure difference across the domain using Dirichlet boundary

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conditions using pflotran (Lichtner et al., 2015), which uses a two-point flux finite volume method to ensure local mass/volume conservation within each mesh control volume. The pressure gradient in many subsurface scenarios of flows through low-permeability rock is typically so small that the flow can be considered laminar (Adler & Thovert, 1999; Bear, 1988; National Research Council, 1996; Nordqvist et al., 2008; The National Academies of Sciences, Engineering, and Medicine, 2021). Recent work has indicated that at higher Reynolds numbers these inertial effects can become important at the single fracture scale (Boutt et al., 2006; Cardenas et al., 2007; Detwiler et al., 2000; Kang et al., 2016), but the incorporation of their inertial effects into network-scale simulations to determine their relative importance at the network scale has yet to be performed and is not addressed here. It is well documented that the classical cubic law, where an effective aperture is used to model flow that replaces two rough surfaces by a parallel plate equivalent model, can be an inaccurate representation of the flow between two rough walled fractures, and especially when the level of asperities is quite high, even when the flow is laminar (Adler & Thovert, 1999; Brush & Thomson, 2003; Thompson & Brown, 1991; Witherspoon et al., 1980). Using the statistical measurements of the aperture field reported above we have confirmed that a local cubic law would be acceptable in these simulations (based on the work of Brush and Thomson [2003]) so long as the Reynolds number is less than one, the latter of which can be ensured by the choice of pressure gradient. We adopt a numerical method where the spatially variable mesh within each fracture plane represents the local aperture variability rather than an integrated value for each mesh cell, details provided in the Supporting Information S1. Solute transport is simulated using a particle tracking approach where a plume of indivisible purely advective particles (no molecular diffusion) trace pathlines through the steady-state fluid velocity field within the network (Hyman, Rajaram, et al., 2019; Makedonska et al., 2015). One-hundred thousand particles are inserted into each network; increasing the number of particles beyond this value did not change the results. The initial location of particles are assigned using flux-weighting (Frampton & Cvetkovic, 2009; Hyman, Painter, et al., 2015; Kang et al., 2017, 2020; Kreft & Zuber, 1978) and a complete mixing rule is used to determine particle behavior at fracture intersections (Berkowitz et al., 1994; Kang et al., 2015; Park et al., 2001, 2003, 2003; Sherman et al., 2019; Stockman et al., 1997). Additional details of the flow and transport simulations are provided in Supporting Information S1.

3. Results

Within each network, we observe the following flow and transport properties: (1) the sequence of fractures that each particle passes through, which are used to construct a flow topology graph (FTG) (Aldrich et al., 2017); (2) the distribution of first passage times of particles from inlet to outlet, the breakthrough curve (BTC); (3) particle tortuosities; (4) the active surface area of flow within the networks; and (5) the degree of flow channelization relative to the constant aperture reference case. Formal definitions of 1–5 are provided in the Supporting Information S1.

A selection of FTGs from one network but with different fracture apertures distributions is shown in Figure 3; the images presented here are representative of all 110 samples considered. Each graph represents a DFN where nodes correspond to fractures and if two fractures intersect there is an edge in the graph between the corresponding nodes (Hyman et al., 2018; Huseby et al., 1997). Node size and edge thickness are proportional to the percentage of particles passing through the corresponding fractures and intersections. Recall that particles are passive tracers and represent a small volume of fluid. Thus, the thickness of the lines provides a visual perspective of fluid volume passing between fractures. The FTG for the constant aperture network is included in the upper left corner for reference, where a handful of primary channels but no dominant channel can be observed. Figure 3a shows that the network-scale structure imposes constraints on where fluid can flow within the network, and results in network-scale channelization even in the absence of in-fracture variability, as has been reported in previous studies of flow in both 2D and 3D fracture network simulations (Bogdanov et al., 2007; de Dreuzy et al., 2001, 2002; 2004, 2012; Frampton & Cvetkovic, 2010; Huseby et al., 2001; Hyman, Aldrich, et al., 2016; Hyman, Dentz, et al., 2019; Hyman & Jiménez-Martínez, 2018; Joyce et al., 2014; Sherman et al., 2020; Sweeney & Hyman, 2020; Tsang & Neretnieks, 1998; Wellman et al., 2009). The variable aperture FTGs each exhibit different flow structures from the reference case. In some cases, a pathway with less resistance occurs and most fluid passes through a new primary sub-network of fractures. In other cases, small aperture values create a region with higher

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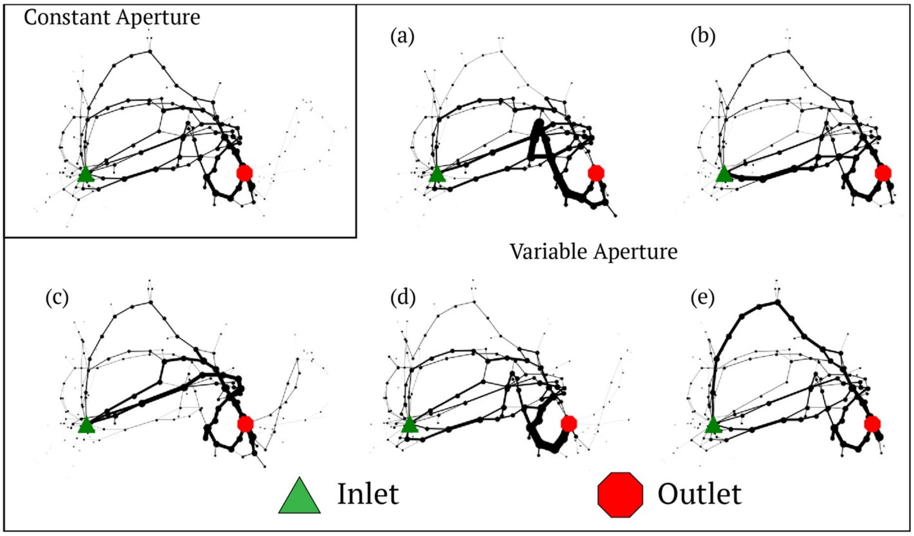

Figure 3. A selection of flow topology graphs (FTG) from one network. Node size and edge thickness are proportional to the percentage of particles passing through the corresponding fractures and intersections. The green triangle denotes the inlet face, and the red octagon denotes the outlet face. The FTG shown in a-e are the same network as the constant aperture network (a), but have different internal aperture fields. Inclusion of internal aperture variability restructures the network-scale flow field and the location of primary flow channels.

resistance close to the inlet and disperses the flow into other regions. The primary channel in sample (a), indicated by being the thickest line in the graph, was a low flow path in the constant network. In sample (b), a single primary channel that connects the inlet to the outlet is observed, and relatively little flow occurs in the rest of the network. In sample (c), there is not a single primary channel, but two. This network exhibited a distribution of tortuosities with a mean value less than observed in the constant case. The flow is relatively dispersed in sample (d), and there is no dominant channel close to the inlet. However, the flow does converge close to the outlet. This network exhibited the highest mean tortuosity in this network set. In sample (e), the primary channels become a rather long path, shown at the top of the sub-figure, because small aperture values block the other primary channels options. Given that particles are passive tracers/packets representing a small volume of fluid, these changes are a Lagrangian perspective showing the reorganization of the Eulerian flow field. These observations demonstrate that local aperture variability within individual fractures can have a scale-bridging effect and modify the network-scale flow field organization.

These rearrangements in the network-scale flow field are manifested in other flow and transport measurements we consider. Adopted definitions/equations for each of the metrics we consider below are provided in the Supporting Information S1. Figure 4a shows the breakthrough curves for the constant networks (blue) and the variable aperture networks (orange). The thick solid line is the mean of the sample realizations, and the shaded area surrounding it indicates the 95% confidence intervals of the ensembles determined using the 10 DFN realizations each with 10 randomly sampled aperture fields mapped onto the fractures; that is, a total number of 110 simulations. Time is normalized by the first arrival time of particles in each of the constant networks (the first arrival time of particles in every constant network is equal to one) to allow for comparison across the ensemble and remove network-to-network based differences. The breakthrough curves (denoted $\psi(\tau)$ where τ is dimensionless time) in both network sets display power-law tailing behavior $\psi(\tau) \sim \tau^{-\alpha+1}$. Compared to the BTC of the constant networks, the first arrival times of particles through the variable networks is consistently earlier, the peak of the BTC is lower, and the decay rate of the tail of the distribution is slower, $\psi(\tau) \propto \tau^{-2.3}$ versus $\psi(\tau) \propto \tau^{-2.6}$ where τ denotes normalized time. Particles pass through more low aperture regions in the variable aperture networks. This results in the change of decay rate. Moreover, the confidence intervals do not overlap, which indicates that these differences cannot be attributed to variations between networks. In combination, these observations indicate significantly larger dispersion of the particle plume in the variable aperture networks resulting from both network-scale

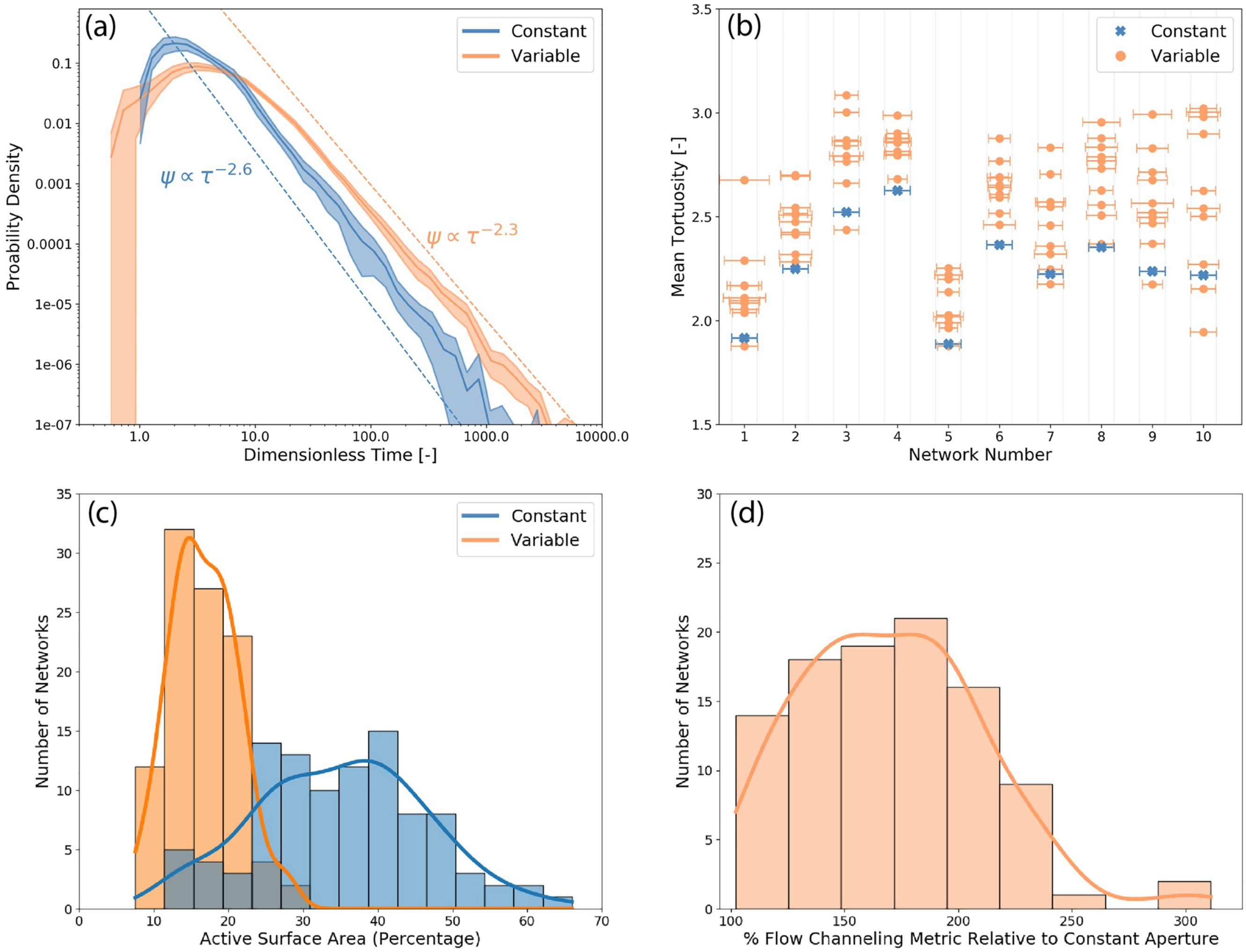

Figure 4. Flow and transport metrics comparing the constant aperture (blue) with the variable aperture (orange) networks. (a) Particle travel time distribution (Breakthrough Curve). (b) Mean and normalized variances of the particle tortuosities for each network sample, which are plotted along the abscissa. (c) Distribution of active surface area. (d) Degree of flow channelization relative to the constant aperture reference case.

heterogeneity and variable aperture. While aperture variability is expected to increase local dispersion, we observe that it can dominant over network structure in terms of network-scale dispersion.

Figure 4b shows the mean and normalized variances of the particle tortuosities for each network realization, which are plotted along the abscissa. We adopt a geometric definition of tortuosity being the total pathline distance over the linear distance between inflow and outflow boundaries. The mean of the constant networks is denoted by a blue cross and the mean of each variable network sample by an orange circle. The variances are normalized by the variance of the constant aperture case for comparison and are shown as error bars. In general, the mean tortuosity of the variable networks is consistently larger than that observed in the reference network; however, there are seven out of the 100 samples that are an exception. These observations indicate the average pathline length in the variable aperture cases are longer than those in the constant aperture case. This trend of increasing pathline length, in part at least, reflects variations within each fracture plane, as has been observed in single fracture studies. Figures 2d and 2e show pathlines in the same network for the constant aperture and the variable aperture case demonstrating this is case within networks as well. The local variations in the aperture result in locally more tortuous pathlines when compared to the constant aperture case. However, these slight variations within the planes cannot account for the cases where the mean pathline length increases by 50% or more, as is observed in several networks.

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Rather, these increases are the result of alterations in the network-scale flow field, as seen in changing FTG structures of Figure 3.

Figure 4c reports the distribution of active surface area from the ensemble defined as the ratio of surface area where there is significant flow over total surface area (Hyman et al., 2020; Maillot et al., 2016). We generated an additional 90 networks with constant apertures so the number of samples in the two distributions (constant and variable) is the same. The distribution of active surface area values from the constant networks is rather broad and reflects network to network variability. In contrast, the distribution of active surface area values from the variable networks is much tighter with consistently lower values. There are marked decreases in the total percentage of the surface area where there is significant flow compared to the constant networks. In most cases the active surface area decreased by over 50%. While some variations in active surface area are expected within a single fracture due to aperture variability, as is known from single-fracture experiments and simulations, the magnitude of these changes reflects the previously observed re-organization of the network-scale flow structure within the network. Large sub-networks and sets of entire fractures where there was a significant amount of flow, contain little to no-flow in the variable aperture networks. These features of flow reorganization can also be seen in the changing edge thickness within Figure 3.

All of these attributes and observations point to increased network-scale flow channelization due to aperture variability. We measure the degree of flow channelization using the method developed in Hyman (2020), where the flow along primary pathways are identified using particle transport, and then ranked to provide a single value for the total mass transported along the primary channels in each network. Figure 4d shows the distribution of the total mass transported along the 20 primary flow channels in the variable networks normalized by the total mass transported along the 20 primary flow channels in the corresponding constant aperture network (values equal to 100% indicate no change; preliminary investigations determined the majority of flow was captured in the first 20 primary channels in these set of simulations.) In every variable aperture sample, the total mass transported along the primary channels is larger than in the reference constant network, indicating an increase in flow channelization. The increase is 170% on average and has a maximum of 311%. Note, as well, that these paths need not be the same sequence of fractures in different realizations of the aperture field in the same network, as can be seen in Figure 3. In summation, these measurements demonstrate that the degree of flow channelization, a network-scale flow structure, is drastically influenced by local changes in the aperture field, which in turn affects flow and transport properties.

4. Discussion and Conclusions

By integrating laboratory observations of fracture apertures formed at high-stress conditions into high-fidelity 3D DFN simulations we identified a novel feature of scale-bridging in fracture networks where micro-scale geologic features can alter macro-scale flow properties. Previous work has shown that the large-scale structure of a fracture network, specifically network geometry and topology, plays a crucial role in the organization of the flow field and formation of primary flow channels. This work lays a foundation for studying flow alteration at the network scale by demonstrating the impact that variations in fracture aperture can have on network-scale transport processes. We observed that the small-scale features of local aperture variability can not only alter primary flow channels within individual fractures but at larger scales as well. These impacts were observed via particle trajectories, active surface area, and the degree of flow channelization.

The results presented provide the beginning of a new understanding of the interplay and hierarchy of length-scales in fractured media and their impact on flow properties. Research in this direction will provide guidelines for when aperture variability is a dominant or ancillary feature. While we focused on steady flow and advective/passive transport in this study, the reported changes in flow observations are likely to influence other geophysical processes, including retention due to matrix diffusion and sorption as well as reactive transport, both fluid-fluid and fluid-solid. Such potential interactions will be investigated in future studies.

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Data Availability Statement

The software used for the simulations, dfnWorks, is open source and can be obtained at http://dfnworks.lanl.gov. Run scripts to generate reproduce data from these simulations are in the Mendeley repository http://dx.doi.org/10.17632/hjdrk5wkfj.1. Los Alamos National Laboratory unclassified release number: LA-UR-21-20,547.

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